Date: 5/Aug/26 Experiment - 5

Packet Capture Tool wireshark

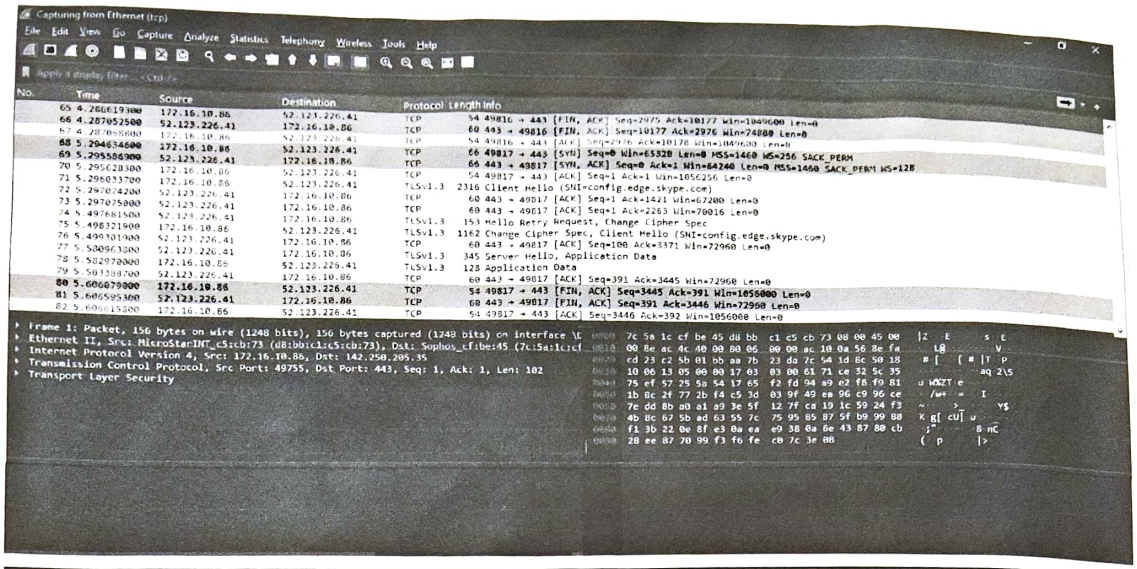
AIM

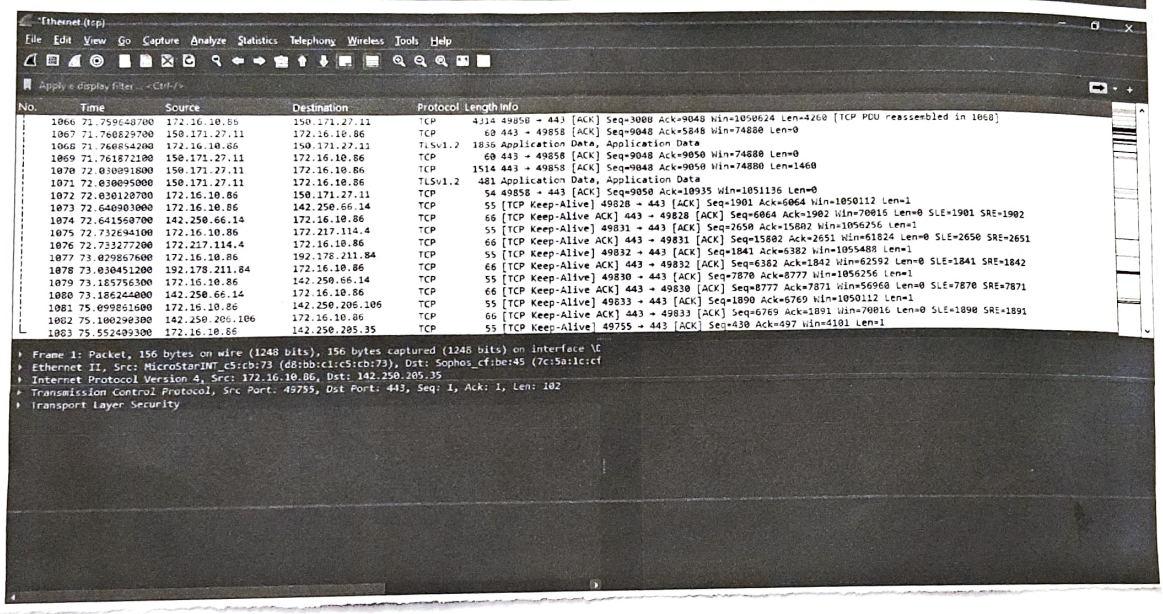
To learn the installation and usage of the wireshark packet sniffer tool to capture, filter, and analyze various network protocols (such as TCP/UDP, ARP, DNS, HTTP, ICMP, and DHCP) and to visualize their communication using flow graphs

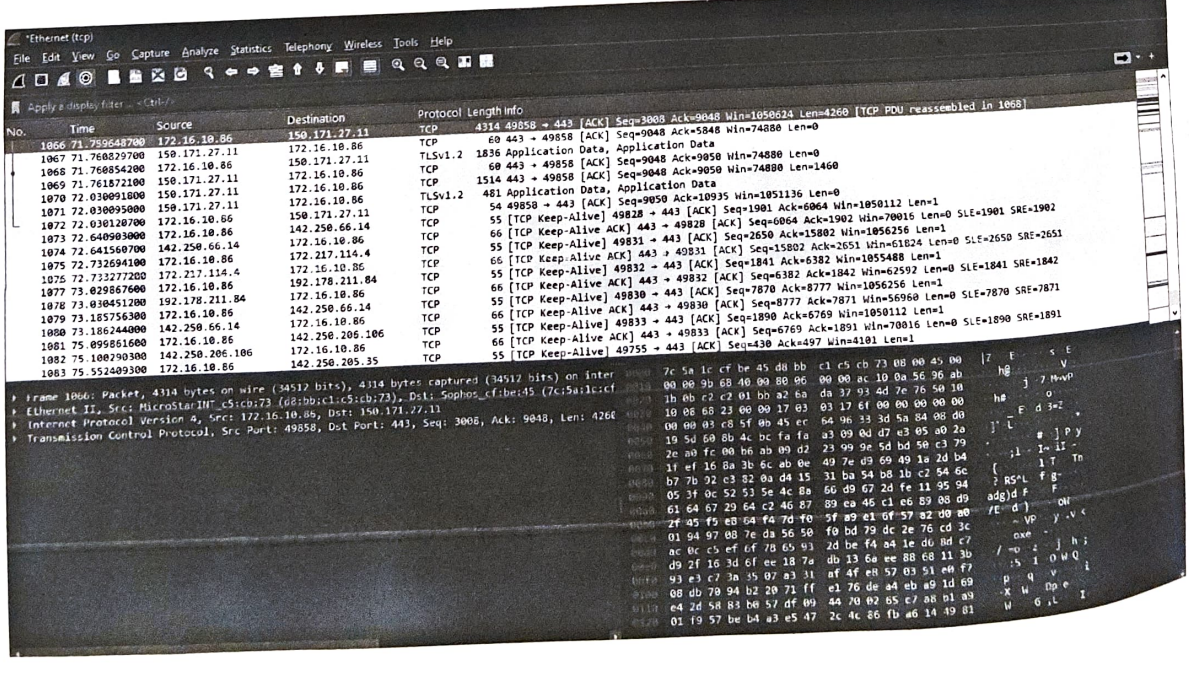
PROCEDURE

Part 1: Initial Setup and Packet Capturing

1. Launch the wireshark application on the system.
2. under the "Capture" menu, select the active network interface (e.g. Local Area Connection / Ethernet).
3. Navigate to Capture → options. verify that "Enable Promiscuous mode on all interfaces" is checked.
4. Set the capture limit by selecting the option to stop capture automatically after 100 packets.
5. click Start to begin capturing network traffic
6. once 100 packets are captured (or by clicking the red "stop" button), save the capture file by navigating to File → Save.

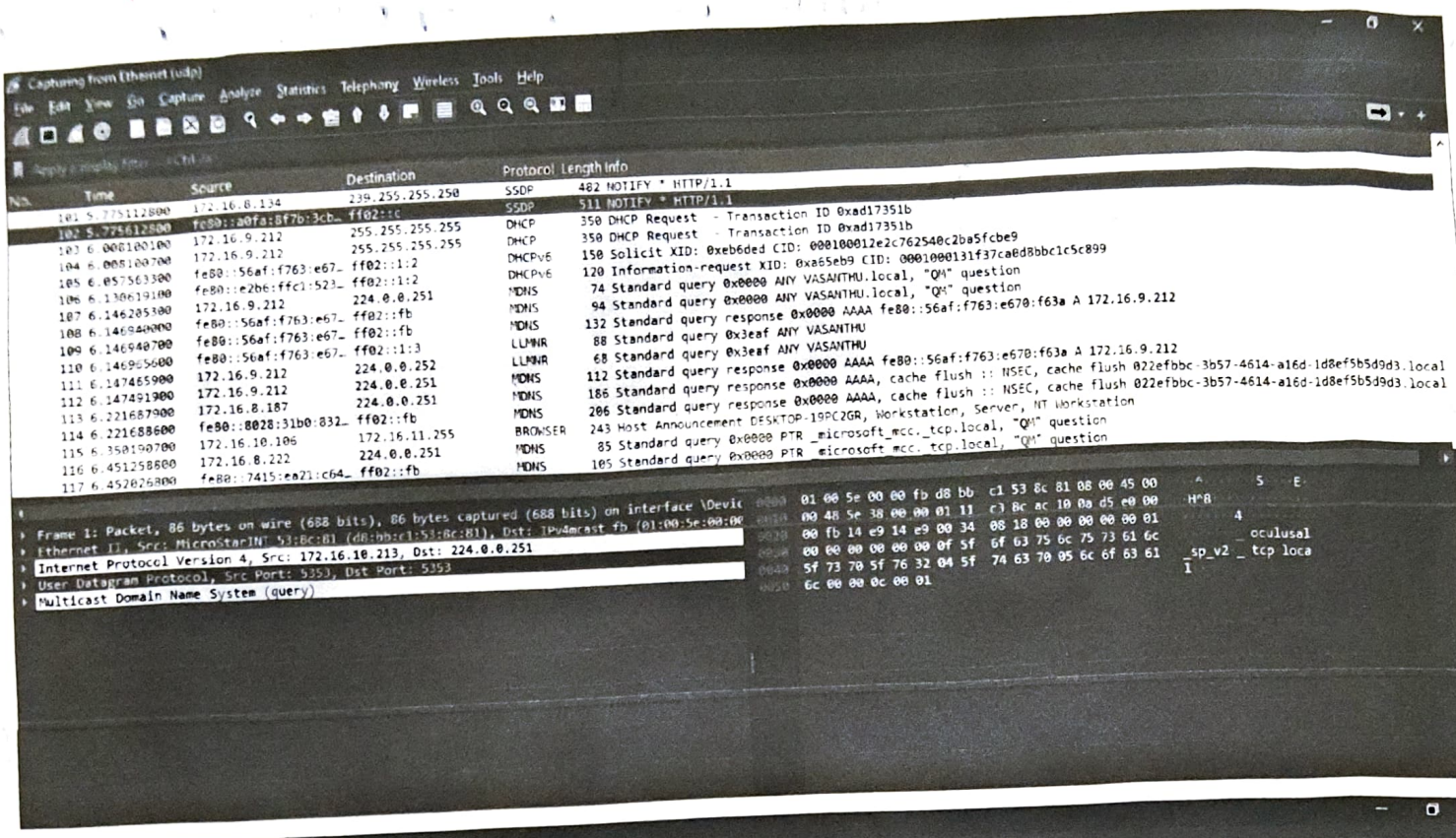

Capturing from Ethernet (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
65 4.286619300 172.16.10.86 52.123.226.41 TCP 54 49816 → 443 [FIN, ACK] Seq=2975 Ack=10177 Win=1049600 Len=0
66 4.287052500 52.123.226.41 172.16.10.86 TCP 60 443 → 49816 [FIN, ACK] Seq=10177 Ack=2976 Win=74880 Len=0
67 4.287086800 172.16.10.86 52.123.226.41 TCP 54 49816 → 443 [ACK] Seq=2976 Ack=10178 Win=1049600 Len=0
68 5.294634600 172.16.10.86 52.123.226.41 TCP 66 49817 → 443 [SYN] Seq=0 Win=65320 Len=0 MSS=1460 WS=256 SACK_PERM
69 5.295586300 52.123.226.41 172.16.10.86 TCP 66 443 → 49817 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
70 5.295628300 172.16.10.86 52.123.226.41 TCP 54 49817 → 443 [ACK] Seq=1 Ack=1 Win=1056256 Len=0
71 5.296033700 172.16.10.86 52.123.226.41 TLSv1.3 2316 Client Hello (SNI=config.edge.skype.com)
72 5.297074200 52.123.226.41 172.16.10.86 TCP 60 443 → 49817 [ACK] Seq=1 Ack=1421 Win=67200 Len=0
73 5.297075000 52.123.226.41 172.16.10.86 TCP 60 443 → 49817 [ACK] Seq=1 Ack=2263 Win=70016 Len=0
74 5.497681500 52.123.226.41 172.16.10.86 TLSv1.3 153 Hello Retry Request, Change Cipher Spec
75 5.498321900 172.16.10.86 52.123.226.41 TLSv1.3 1162 Change Cipher Spec, Client Hello (SNI=config.edge.skype.com)
76 5.499030100 52.123.226.41 172.16.10.86 TCP 60 443 → 49817 [ACK] Seq=100 Ack=3371 Win=72960 Len=0
77 5.500963000 52.123.226.41 172.16.10.86 TLSv1.3 345 Server Hello, Application Data
78 5.502970000 172.16.10.86 52.123.226.41 TLSv1.3 128 Application Data
79 5.503398700 52.123.226.41 172.16.10.86 TCP 60 443 → 49817 [ACK] Seq=391 Ack=3445 Win=72960 Len=0
80 5.606679000 172.16.10.86 52.123.226.41 TCP 54 49817 → 443 [FIN, ACK] Seq=3445 Ack=391 Win=1056000 Len=0
81 5.606595300 52.123.226.41 172.16.10.86 TCP 60 443 → 49817 [FIN, ACK] Seq=391 Ack=3446 Win=72960 Len=0
82 5.606615300 172.16.10.86 52.123.226.41 TCP 54 49817 → 443 [ACK] Seq=3446 Ack=392 Win=1056000 Len=0
Frame 1: Packet, 156 bytes on wire (1248 bits), 156 bytes captured (1248 bits) on interface \E
Ethernet II, Src: MicroStarINT_c5:cb:73 (d8:bb:c1:c5:cb:73), Dst: Sophos_cf:be:45 (7c:5a:1c:cf
Internet Protocol Version 4, Src: 172.16.10.86, Dst: 142.250.205.35
Transmission Control Protocol, Src Port: 49755, Dst Port: 443, Seq: 1, Ack: 1, Len: 102
Transport Layer Security

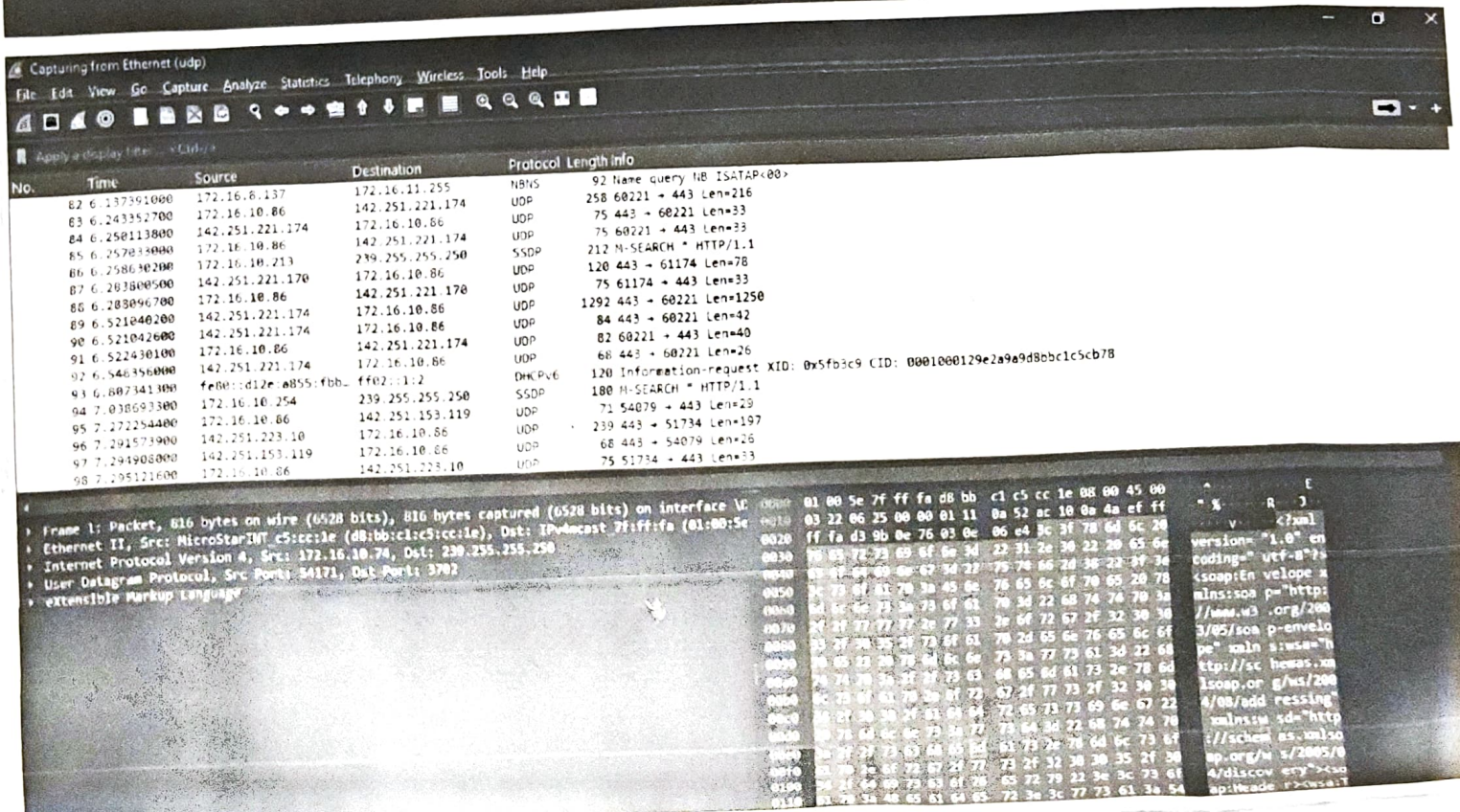

*Ethernet (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
1066 71.759648700 172.16.10.86 150.171.27.11 TCP 4314 49858 → 443 [ACK] Seq=3008 Ack=9048 Win=1050624 Len=4260 [TCP PDU reassembled in 1068]
1067 71.760829700 150.171.27.11 172.16.10.86 TCP 60 443 → 49858 [ACK] Seq=9048 Ack=5848 Win=74880 Len=0
1068 71.760854200 172.16.10.86 150.171.27.11 TLSv1.2 1836 Application Data, Application Data
1069 71.761872100 150.171.27.11 172.16.10.86 TCP 60 443 → 49858 [ACK] Seq=9048 Ack=9050 Win=74880 Len=0
1070 72.030091800 150.171.27.11 172.16.10.86 TCP 1514 443 → 49858 [ACK] Seq=9048 Ack=9050 Win=74880 Len=1460
1071 72.030095000 150.171.27.11 172.16.10.86 TLSv1.2 481 Application Data, Application Data
1072 72.030120700 172.16.10.86 150.171.27.11 TCP 54 49858 → 443 [ACK] Seq=9050 Ack=10935 Win=1051136 Len=0
1073 72.640903000 172.16.10.86 142.250.66.14 TCP 55 [TCP Keep-Alive] 49828 → 443 [ACK] Seq=1901 Ack=6064 Win=1050112 Len=1
1074 72.641560700 142.250.66.14 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49828 [ACK] Seq=6064 Ack=1902 Win=70016 Len=0 SLE=1901 SRE=1902
1075 72.732694100 172.16.10.86 172.217.114.4 TCP 55 [TCP Keep-Alive] 49831 → 443 [ACK] Seq=2650 Ack=15802 Win=1056256 Len=1
1076 72.733277200 172.217.114.4 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49831 [ACK] Seq=15802 Ack=2651 Win=61824 Len=0 SLE=2650 SRE=2651
1077 73.029867600 172.16.10.86 192.178.211.84 TCP 55 [TCP Keep-Alive] 49832 → 443 [ACK] Seq=1841 Ack=6382 Win=1055488 Len=1
1078 73.030451200 192.178.211.84 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49832 [ACK] Seq=6382 Ack=1842 Win=62592 Len=0 SLE=1841 SRE=1842
1079 73.185756300 172.16.10.86 142.250.66.14 TCP 55 [TCP Keep-Alive] 49830 → 443 [ACK] Seq=7870 Ack=8777 Win=1056256 Len=1
1080 73.186244000 142.250.66.14 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49830 [ACK] Seq=8777 Ack=7871 Win=56960 Len=0 SLE=7870 SRE=7871
1081 75.099861600 172.16.10.86 142.250.206.106 TCP 55 [TCP Keep-Alive] 49833 → 443 [ACK] Seq=1890 Ack=6769 Win=1050112 Len=1
1082 75.100290300 142.250.206.106 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49833 [ACK] Seq=6769 Ack=1891 Win=70016 Len=0 SLE=1890 SRE=1891
1083 75.552409300 172.16.10.86 142.250.205.35 TCP 55 [TCP Keep-Alive] 49755 → 443 [ACK] Seq=430 Ack=497 Win=4101 Len=1
Frame 1: Packet, 156 bytes on wire (1248 bits), 156 bytes captured (1248 bits) on interface \E
Ethernet II, Src: MicroStarINT_c5:cb:73 (d8:bb:c1:c5:cb:73), Dst: Sophos_cf:be:45 (7c:5a:1c:cf
Internet Protocol Version 4, Src: 172.16.10.86, Dst: 142.250.205.35
Transmission Control Protocol, Src Port: 49755, Dst Port: 443, Seq: 1, Ack: 1, Len: 102
Transport Layer Security


*Ethernet (tcp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
1066 71.759648700 172.16.10.86 150.171.27.11 TCP 4314 49858 → 443 [ACK] Seq=3008 Ack=9048 Win=1050624 Len=4260 [TCP PDU reassembled in 1068]
1067 71.760829700 150.171.27.11 172.16.10.86 TCP 60 443 → 49858 [ACK] Seq=9048 Ack=5848 Win=74880 Len=0
1068 71.760854200 172.16.10.86 150.171.27.11 TLSv1.2 1836 Application Data, Application Data
1069 71.761872100 150.171.27.11 172.16.10.86 TCP 60 443 → 49858 [ACK] Seq=9048 Ack=9050 Win=74880 Len=0
1070 72.030091800 150.171.27.11 172.16.10.86 TCP 1514 443 → 49858 [ACK] Seq=9048 Ack=9050 Win=74880 Len=1460
1071 72.030095000 150.171.27.11 172.16.10.86 TLSv1.2 481 Application Data, Application Data
1072 72.030120700 172.16.10.86 150.171.27.11 TCP 54 49858 → 443 [ACK] Seq=9050 Ack=10935 Win=1051136 Len=0
1073 72.640903000 172.16.10.86 142.250.66.14 TCP 55 [TCP Keep-Alive] 49828 → 443 [ACK] Seq=1901 Ack=6064 Win=1050112 Len=1
1074 72.641560700 142.250.66.14 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49828 [ACK] Seq=6064 Ack=1902 Win=70016 Len=0 SLE=1901 SRE=1902
1075 72.732694100 172.16.10.86 172.217.114.4 TCP 55 [TCP Keep-Alive] 49831 → 443 [ACK] Seq=2650 Ack=15802 Win=1056256 Len=1
1076 72.733277200 172.217.114.4 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49831 [ACK] Seq=15802 Ack=2651 Win=61824 Len=0 SLE=2650 SRE=2651
1077 73.029867600 172.16.10.86 192.178.211.84 TCP 55 [TCP Keep-Alive] 49832 → 443 [ACK] Seq=1841 Ack=6382 Win=1055488 Len=1
1078 73.030451200 192.178.211.84 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49832 [ACK] Seq=6382 Ack=1842 Win=62592 Len=0 SLE=1841 SRE=1842
1079 73.185756300 172.16.10.86 142.250.66.14 TCP 55 [TCP Keep-Alive] 49830 → 443 [ACK] Seq=7870 Ack=8777 Win=1056256 Len=1
1080 73.186244000 142.250.66.14 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49830 [ACK] Seq=8777 Ack=7871 Win=56960 Len=0 SLE=7870 SRE=7871
1081 75.099861600 172.16.10.86 142.250.206.106 TCP 55 [TCP Keep-Alive] 49833 → 443 [ACK] Seq=1890 Ack=6769 Win=1050112 Len=1
1082 75.100290300 142.250.206.106 172.16.10.86 TCP 66 [TCP Keep-Alive ACK] 443 → 49833 [ACK] Seq=6769 Ack=1891 Win=70016 Len=0 SLE=1890 SRE=1891
1083 75.552409300 172.16.10.86 142.250.205.35 TCP 55 [TCP Keep-Alive] 49755 → 443 [ACK] Seq=430 Ack=497 Win=4101 Len=1
Frame 1066: Packet, 4314 bytes on wire (34512 bits), 4314 bytes captured (34512 bits) on inter
Ethernet II, Src: MicroStarINT_c5:cb:73 (d8:bb:c1:c5:cb:73), Dst: Sophos_cf:be:45 (7c:5a:1c:cf
Internet Protocol Version 4, Src: 172.16.10.86, Dst: 150.171.27.11
Transmission Control Protocol, Src Port: 49858, Dst Port: 443, Seq: 3008, Ack: 9048, Len: 4260

UDP

TCP


Capturing from Ethernet (udp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
101 5.775112800 172.16.8.134 239.255.255.250 SSDP 482 NOTIFY * HTTP/1.1
102 5.775612800 fe80::a0fa:8f7b:3cb… ff02::c SSDP 511 NOTIFY * HTTP/1.1
103 6.008100100 172.16.9.212 255.255.255.255 DHCP 350 DHCP Request - Transaction ID 0xad17351b
104 6.008100700 172.16.9.212 255.255.255.255 DHCP 350 DHCP Request - Transaction ID 0xad17351b
105 6.057563300 fe80::56af:f763:e67… ff02::1:2 DHCPv6 150 Solicit XID: 0xeb6ded CID: 000100012e2c762540c2ba5fcbe9
106 6.130619100 fe80::e2b6:ffc1:523… ff02::1:2 DHCPv6 120 Information-request XID: 0xa65eb9 CID: 000100011f37ca0d8bbc1c5c899
107 6.146205300 172.16.9.212 224.0.0.251 MDNS 74 Standard query 0x0000 ANY VASANTHU.local, "QM" question
108 6.146940000 fe80::56af:f763:e67… ff02::fb MDNS 94 Standard query 0x0000 ANY VASANTHU.local, "QM" question
109 6.146940700 fe80::56af:f763:e67… ff02::fb MDNS 132 Standard query response 0x0000 AAAA fe80::56af:f763:e670:f63a A 172.16.9.212
110 6.146965600 fe80::56af:f763:e67… ff02::1:3 LLMNR 88 Standard query 0x3eaf ANY VASANTHU
111 6.147465900 172.16.9.212 224.0.0.252 LLMNR 68 Standard query 0x3eaf ANY VASANTHU
112 6.147491900 172.16.9.212 224.0.0.251 MDNS 112 Standard query response 0x0000 AAAA fe80::56af:f763:e670:f63a A 172.16.9.212
113 6.221687900 172.16.8.187 224.0.0.251 MDNS 186 Standard query response 0x0000 AAAA, cache flush :: NSEC, cache flush 022efbbc-3b57-4614-a16d-1d8ef5b5d9d3.local
114 6.221688600 fe80::8028:31b0:832… ff02::fb MDNS 206 Standard query response 0x0000 AAAA, cache flush :: NSEC, cache flush 022efbbc-3b57-4614-a16d-1d8ef5b5d9d3.local
115 6.350190700 172.16.10.106 172.16.11.255 BROWSER 243 Host Announcement DESKTOP-19PC2GR, Workstation, Server, NT Workstation
116 6.451258800 172.16.8.222 224.0.0.251 MDNS 85 Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
117 6.452026800 fe80::7415:ea21:c64… ff02::fb MDNS 105 Standard query 0x0000 PTR _microsoft_mcc._tcp.local, "QM" question
Frame 1: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Devic
Ethernet II, Src: MicroStarINT_53:6c:81 (d8:bb:c1:53:6c:81), Dst: IPv4mcast_fb (01:00:5e:00:00
Internet Protocol Version 4, Src: 172.16.10.213, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
0000 01 00 5e 00 00 fb d8 bb c1 53 6c 81 08 00 45 00
0010 00 48 5e 38 00 00 01 11 c3 8c ac 10 0a d5 e0 00
0020 00 fb 14 e9 14 e9 00 34 08 18 00 00 00 00 00 01
0030 00 00 00 00 00 00 0f 5f 6f 63 75 6c 75 73 61 6c
0040 5f 73 70 5f 76 32 04 5f 74 63 70 05 6c 6f 63 61
0050 6c 00 00 0c 00 01


Capturing from Ethernet (udp)
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
82 6.137391000 172.16.8.137 172.16.11.255 NBNS 92 Name query NB ISATAP<00>
83 6.243352700 172.16.10.86 142.251.221.174 UDP 258 60221 → 443 Len=216
84 6.250113800 142.251.221.174 172.16.10.86 UDP 75 443 → 60221 Len=33
85 6.257033000 172.16.10.86 142.251.221.174 UDP 75 60221 → 443 Len=33
86 6.258630200 172.16.10.213 239.255.255.250 SSDP 212 M-SEARCH * HTTP/1.1
87 6.283800500 142.251.221.170 172.16.10.86 UDP 120 443 → 61174 Len=78
88 6.288096700 172.16.10.86 142.251.221.170 UDP 75 61174 → 443 Len=33
89 6.521040200 142.251.221.174 172.16.10.86 UDP 1292 443 → 60221 Len=1250
90 6.521042600 142.251.221.174 172.16.10.86 UDP 84 443 → 60221 Len=42
91 6.522430100 172.16.10.86 142.251.221.174 UDP 82 60221 → 443 Len=40
92 6.546356000 142.251.221.174 172.16.10.86 UDP 68 443 → 60221 Len=26
93 6.807341300 fe80::d12e:e855:fbb… ff02::1:2 DHCPv6 120 Information-request XID: 0x5fb3c9 CID: 0001000129e2a9a9d8bbc1c5cb78
94 7.038693300 172.16.10.254 239.255.255.250 SSDP 180 M-SEARCH * HTTP/1.1
95 7.272254400 172.16.10.86 142.251.153.119 UDP 71 54079 → 443 Len=29
96 7.291573900 142.251.223.10 172.16.10.86 UDP 239 443 → 51734 Len=197
97 7.294908800 142.251.153.119 172.16.10.86 UDP 68 443 → 54079 Len=26
98 7.295121600 172.16.10.86 142.251.223.10 UDP 75 51734 → 443 Len=33
Frame 1: Packet, 816 bytes on wire (6528 bits), 816 bytes captured (6528 bits) on interface \D
Ethernet II, Src: MicroStarINT_c5:cc:1e (d8:bb:c1:c5:cc:1e), Dst: IPv6mcast_7f:ff:fa (01:00:5e
Internet Protocol Version 4, Src: 172.16.10.74, Dst: 239.255.255.250
User Datagram Protocol, Src Port: 54171, Dst Port: 3702
eXtensible Markup Language
0000 01 00 5e 7f ff fa d8 bb c1 c5 cc 1e 08 00 45 00
0010 03 22 06 25 00 00 01 11 0a 52 ac 10 0a 4a ef ff
0020 ff fa d3 9b 0e 76 03 0e 06 e4 3c 3f 78 6d 6c 20
0030 76 65 72 73 69 6f 6e 3d 22 31 2e 30 22 20 65 6e
0040 63 6f 64 69 6e 67 3d 22 75 74 66 2d 38 22 3f 3e
0050 3c 73 3a 45 6e 76 65 6c 6f 70 65 20 78 6d 6c 6e
0060 73 3a 73 6f 61 70 3d 22 68 74 74 70 3a 2f 2f 77
0070 77 77 2e 77 33 2e 6f 72 67 2f 32 30 30 33 2f 30
0080 35 2f 73 6f 61 70 2d 65 6e 76 65 6c 6f 70 65 22
0090 20 78 6d 6c 6e 73 3a 77 73 61 3d 22 68 74 74 70
00a0 3a 2f 2f 73 63 68 65 6d 61 73 2e 78 6d 6c 73 6f
00b0 61 70 2e 6f 72 67 2f 77 73 2f 32 30 30 35 2f 30
00c0 34 2f 61 64 64 72 65 73 73 69 6e 67 22 20 78 6d
00d0 6c 6e 73 3a 77 73 64 3d 22 68 74 74 70 3a 2f 2f
00e0 73 63 68 65 6d 61 73 2e 78 6d 6c 73 6f 61 70 2e
00f0 6f 72 67 2f 77 73 2f 32 30 30 35 2f 30 34 2f 64
0100 69 73 63 6f 76 65 72 79 22 3e 3c 77 73 61 3a 54

Part 2: Applying Filters and Analyzing protocols

Use the filter bar at the top of the wireshark window to isolate and inspect specific types of traffic:

1. TCP/UDP Analysis: Type tcp or udp in the filter bar and press Enter. Inspect the packets. To view the flow of data, click statistics → Flow graph.

2. ARP Analysis: Type arp in the filter bar. Select an ARP packet and expand the "packet Details" pane to inspect the sender and

Result:

Thus the packet capture Tool wireshark to learn installation and usage is done successfully